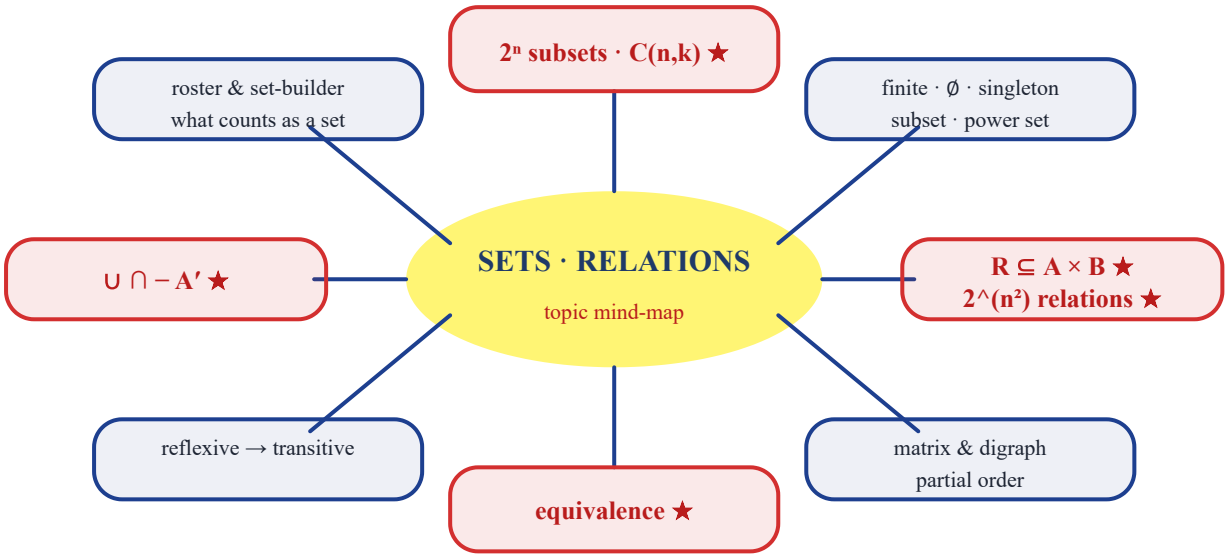
Set Theory, Relations & Functions

2^n subsets · set laws · relations $\subseteq A \times B$ · injective / onto: exam-ready in 11 sheets



↑ the topic on one page: red = high-yield, blue = definitions

Syllabus coverage

"Discrete Mathematics, Theory of Computation and Basic Graph Theory: **Set theory**; **Relations**; **Functions**; Graph theory; Propositional calculus and predicate calculus; …" This notebook covers the first three; graph theory and predicate calculus have their own lessons in the chapter.

High-yield list

① 4 elements ⇒ $2^4 = 16$ subsets
② 2^n subsets; C(n,k) of size k
③ $|A \cup B| = |A| + |B| - |A \cap B|$
④ De Morgan: $(A \cup B)' = A' \cap B'$
⑤ a relation = **subset** of $A \times B$
⑥ 3 elements ⇒ $2^9 = 512$ relations
⑦ injective ⇒ $|A| \le |B|$; onto ⇒ $\ge$
⑧ A→B has |B|^|A| functions
⑨ $(f \circ g)(x) = f(g(x))$: f **after** g
⑩ bijective ⇔ f^{-1} exists

Legend: **yellow** = key term · **red circle** = must-memorise fact · **purple box** = memory trick · **red box** = exam trap · **green** = worked example

Contents

✓ What a set is; the two forms … p. 2
✓ Counting: 2^n and C(n,k) … p. 2
✓ Types of sets & power set … p. 3
✓ Set operations & the laws … p. 4
✓ Relations $\subseteq A \times B$ … p. 5
✓ Six properties, counter-examples … p. 6
✓ Equivalence & partial order … p. 7
✓ Functions & one-to-one / onto … p. 8
✓ Composition, inverse, counting … p. 9
✓ Night-before p. 10 · Practice p. 11

1 What Is a Set?

Definition: A **set** is a **well-defined collection of distinct objects**, its **elements** or **members**. "Well-defined" means we can decide unambiguously **yes or no** whether an object belongs. The objects may be numbers, letters, points, or even other sets.

Form	It does what	Example
Roster (tabular)	lists **every** element one by one	A = {2, 4, 6, 8}
Set-builder	states the **property** the elements share	A = {x | x even, x < 10}

Order and repetition are irrelevant: {1, 2, 3} = {3, 2, 1} = {1, 1, 2, 2} · but (1, 2) ≠ (2, 1) — **ordered pairs do care**
|A| is read "the **cardinality** of A": |{a, b}| = 2.

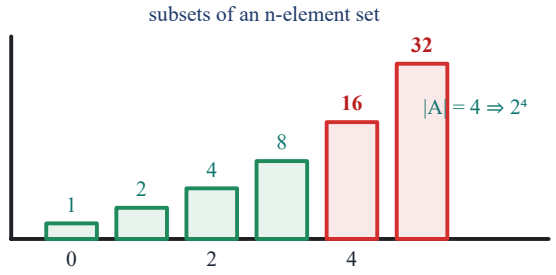
Exam trap: **{even numbers}** is **not** a set — it is not well defined until you say which evens. That clause is the whole definition and it is asked often.

Memory trick: A set is a **basket with a label**: the label decides who is in, and the order of picking never changes the contents.

2 Counting Subsets: 2^n and C(n,k)

The counting argument: Walk through the **n** elements one at a time; each has exactly **two choices — put it in, or leave it out**. Two choices, n times, gives $\mathbf{2^n}$ subsets, counting both **∅ and A itself**.

n	$\|P(A)\| = 2^n$	of size 2	proper
0 (A = ∅)	**1**	0	0
2	4	1	3
3	8	3	7
4	**16**	**6**	**15**
5	32	10	31



every extra element **doubles** the count — 2^n grows faster than anything else here

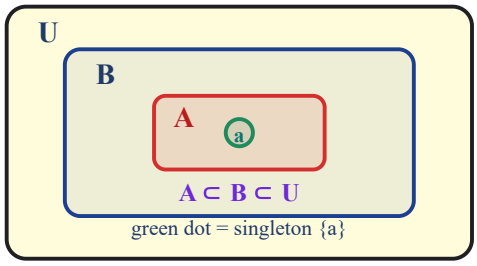
Worked example ✍️: "A set A has **4 elements**. How many subsets?" → **16**, not 8 and not 4. **Proper** subsets → **15**; excluding the empty set too → **14**. Subsets of size k = C(n,k) = n!/k!(n−k)!; for n = 4: 4 + 6 + 4 + 1 = 15, and adding ∅ gives 16 ✓

3 Types of Sets, Subsets & the Power Set

Name	Definition	Example
Finite / infinite	a definite countable number of elements, $\|A\| = n$ / elements never run out	$\{1, 2, 3\}$ / $\mathbb{N}, \mathbb{Z}, \mathbb{Q}, \mathbb{R}$
Null / empty set $\emptyset$	**no** elements at all; also written $\{\}$ or φ	$\emptyset = \{x \mid 1 = 0\}$
Singleton set	**exactly one** element	$\{a\}$
Equal sets	the same elements, in any order: $A = B$	$\{1, 2, 3\} = \{3, 2, 1\}$
Subset $A \subseteq B$	every element of A is in B	$\{1, 2\} \subseteq \{1, 2, 3\}$
Proper subset $A \subset B$	$A \subseteq B$ **and** $A \neq B$	$\{1, 2\} \subset \{1, 2, 3\}$
Universal set U	every element under discussion belongs to it	$U = \{1, 2, 3, 4, 5, 6\}$
Complement A'	everything in U that is **not** in A: $A' = U - A$	$U = \{1,2,3,4\}, A = \{1,2\} \Rightarrow \{3,4\}$

Power set: **P(A)** = $\{ X \mid X \subseteq A \}$ — the set of **all** subsets of A. Then $|\mathbf{P(A)}| = \mathbf{2^{|A|}}$ and, for every finite A, $|\mathbf{P(A)}| > |\mathbf{A}|$ (Cantor). Edge cases: $|P(\emptyset)| = 1$ and $P(\{a\}) = \{ \emptyset, \{a\} \}$, so P(A) itself is a set and P(P(A)) exists too.

Memory trick: **$\subseteq$ is "all in", $\subset$ is "all in and more outside"**. Every set is a subset of itself ($\mathbf{A \subseteq A}$) and $\emptyset$ is a subset of every set ($\mathbf{\emptyset \subseteq A}$): $\emptyset$ is the **smallest** subset, U the **largest**.



nested the other way round reads as **superset**: $B \supset A$ means B has everything A has, and more

Worked example ✍️ subsets of A = {1, 2, 3, 4}: total 16 · exactly 2 elements → 6 · exactly 3 → 4 · proper → 15 · proper and non-empty → 14. Since $|P(A)| = 16 > 4$, $A \subset P(A)$ strictly, and the largest subset is A itself.

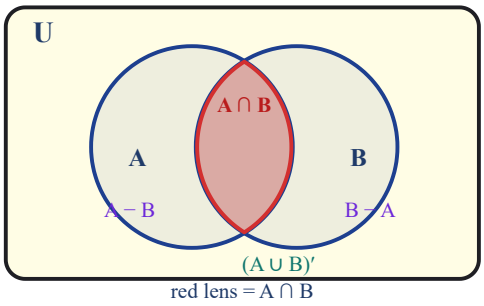
Exam trap: (i) $\emptyset \subset \{x\}$ holds, but $\{x\} \subset \emptyset$ is **false** — $\emptyset$ has no elements to miss. (ii) The empty set is **not** equal to $\{0\}$ or $\{\emptyset\}$. (iii) "Subset" never means "proper subset" in ordinary exam wording.

IT link: The power set is not an exam abstraction: an **n-bit flag word has exactly 2^n settings**, and asking "which of these 32 permissions is on?" is a power-set question in disguise. One bit per permission, every combination legal, nothing missing. The same count gives the number of **subsets a table can filter on**, which is why a query with n boolean predicates has at most 2^n distinct result shapes.

4 Set Operations and the Laws of Set Algebra

Operation	Symbol & definition	Worked instance
Union	$A \cup B = \{x \mid x \in A$ **or** $x \in B\}$ — everything in either	$\{1,2\} \cup \{2,3\} = \{1,2,3\}$
Intersection	$A \cap B = \{x \mid x \in A$ **and** $x \in B\}$ — the common part	$\{1,2\} \cap \{2,3\} = \{2\}$
Difference	$A - B = \{x \mid x \in A$ and $x \notin B\}$ — A with B removed	$\{1,2,3\} - \{2,3\} = \{1\}$
Complement	$A' = \{x \mid x \in U$ and $x \notin A\}$ — $U - A$	$U = \{1,2,3,4\}$, $A = \{1,2\} \Rightarrow \{3,4\}$
Cartesian product	$A \times B = \{(a, b) \mid a \in A, b \in B\}$ — all ordered pairs	$\{1,2\} \times \{x,y\} = \{(1,x),(1,y),(2,x),(2,y)\}$

Law	Statement
Commutative	$A \cup B = B \cup A \cdot A \cap B = B \cap A$
Associative	$(A \cup B) \cup C = A \cup (B \cup C)$ · same for $\cap$
Distributive	$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$
Identity	$A \cup \emptyset = A \cdot A \cap U = A$
Complement	$A \cup A' = U \cdot A \cap A' = \emptyset$
Idempotent	$A \cup A = A \cdot A \cap A = A$
Involution	$(A')' = A$
Absorption	$A \cup (A \cap B) = A \cdot A \cap (A \cup B) = A$
De Morgan	$\mathbf{(A \cup B)' = A' \cap B' \cdot (A \cap B)' = A' \cup B'}$



red lens = $A \cap B$

only the overlap is the intersection; everything inside **both circles** is the union

Cardinality formulas you must know:
$|A \cup B| = |A| + |B| - |A \cap B|$
If $A \cap B = \emptyset$ (**disjoint**) this collapses to $|A| + |B|$. Generalising to three sets gives inclusion–exclusion, which is a separate lesson.

Exam trap: $A - B$ and $B - A$ are **different sets**, so difference is **not** commutative: $\{1,2,3\} - \{2,3\} = \{1\}$ but $\{2,3\} - \{1,2,3\} = \emptyset$. And $A \times B \neq B \times A$ as sets, though $|A \times B| = |B \times A|$.

Worked example ✍ simplify with the laws: $(A \cup B)' = A' \cap B'$ by De Morgan, and $((A \cup B)')' = A' \cap B'$ returns to $A \cup B$ by **involution**. Also $A - B = A \cap B'$, so $\{1,2,3\} - \{2,3\} = \{1,2,3\} \cap \{1\} = \{1\}$. De Morgan · involution · difference in terms of intersection and complement

5 A Relation Is a Subset of A × B

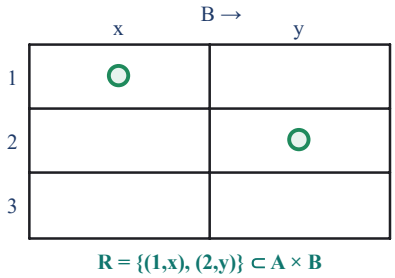
Definition: Let A and B be two sets. A **relation R from A to B** is **any subset of A × B**, written $R \subseteq A \times B$. Its elements are **ordered pairs** (a, b) with $a \in A$, $b \in B$, read "a is related to b". If $R \subseteq A \times A$ it is a **binary relation on A**; that is the case whenever both sets are the same.

The two counting results: $|A \times B| = |A| \cdot |B|$ — the pairs form a full grid, m rows by n columns. **Relations $A \to B = 2^{(|A| \cdot |B|)}$**, because a relation is **any** subset and a set of $m \cdot n$ elements has $2^{(m \cdot n)}$ subsets. On a single set with n elements: $|A \times A| = n^2$, so **$2^{(n^2)}$ binary relations**.

Worked example ✍️ the counts that always appear: **(a)** $|A| = 3$: $|A \times A| = 9$, so $\mathbf{2^9 = 512}$ relations on A. **(b)** $|A| = 3$, $|B| = 4$: $|A \times B| = 12$, so $2^{12} = \mathbf{4096}$ relations $A \to B$. **(c)** $|A| = 4$: $\mathbf{2^{16} = 65\,536}$. The count explodes because the exponent is n^2, not n.

Exam trap: **Not every relation is a function.** A relation may contain (1, x) **and** (1, y) for the same first element; a function may leave part of the codomain untouched. And $R \subseteq A \times B$ does **not** mean $R = A \times B$ — the universal relation is only one of the $2^{(m \cdot n)}$ choices.

IT link: A database join condition is a **relation** between two tables, and a foreign key declares one.



$R = \{(1,x), (2,y)\} \subset A \times B$

A = {1,2,3}, B = {x,y}: the grid has $|A| \cdot |B| = 6$ cells and a relation may mark **any** subset — $2^6 = 64$ relations $A \to B$

Three ways to write the same relation

1. **Set of ordered pairs** — R = {(1, x), (2, y)}
2. **Matrix** $M_R = [m_{ij}]$, rows = A, columns = B, m_{ij} = 1 if $(a_i, b_j) \in R$, else 0
3. **Digraph** — one vertex per element of A and one per element of B, with a directed edge $a_i \to b_j$ for every pair in R. For a relation **on** A there is a single vertex set, and (a, a) is drawn as a **self-loop**

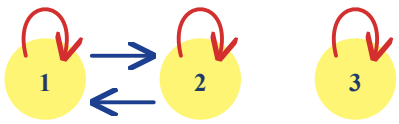
6 The Six Properties, with a Counter-example for Each

Property	Exact statement (on a set A)	Counter-example on A = {1, 2, 3}
Reflexive	$(a, a) \in R$ for **every** $a \in A$	R = {(1,2),(2,3)} — (3,3) is missing
Not reflexive (irreflexive)	$(a, a) \notin R$ for every $a \in A$	R = {(1,1)} on {1,2,3} — (2,2) is present
Symmetric	$(a, b) \in R \Rightarrow (b, a) \in R$	R = {(1,2),(1,1)} — (2,1) is missing
Antisymmetric	$(a, b) \in R$ and $(b, a) \in R \Rightarrow a = b$	R = {(1,2),(2,1)} — both ways, yet $1 \neq 2$
Transitive	$(a, b) \in R$ and $(b, c) \in R \Rightarrow (a, c) \in R$	R = {(1,2),(2,3)} — forces (1,3), missing
Asymmetric	$(a, b) \in R$ and $a \neq b \Rightarrow (b, a) \notin R$	R = {(1,2),(2,1)} — symmetry holds instead

The distinction that loses marks: **Symmetric** and **antisymmetric** are **not opposites** and neither implies the other. R = {(1,2),(2,1)} fails antisymmetry but **is** symmetric. R = {(1,2)} is antisymmetric yet **not** symmetric. Only "asymmetric" is the true opposite of "symmetric".

Memory trick: **Symmetric = the arrow reverses; antisymmetric = at most one arrow per unordered pair.** Two more worth memorising: ∅ on a non-empty A is symmetric, antisymmetric and transitive but **not** reflexive, while A × A is reflexive, symmetric and transitive but **not** antisymmetric. The identity relation {(a,a)} is the only one that passes all four.

Worked example ✍ matrix form: For R = {(1,x), (2,y)} with rows 1,2,3 and columns x,y: M_R = [[1,0],[0,1],[0,0]]. On a relation **on** A rows and columns are the same list, so **the diagonal is the reflexivity test**.



R = {(1,1),(2,2),(3,3),(1,2),(2,1)}
loops everywhere + 1↔2 ⇒ equivalence

reflexivity = a **loop at every** vertex; transitivity = every path of length two has a shortcut arrow

7 Equivalence Relations and Partial Orders

Equivalence relation: **reflexive + symmetric + transitive**, nothing more. Its classes [a] = { x ∈ A | (a, x) ∈ R } are non-empty, pairwise disjoint, and their union is A.
Equivalence relation ↔ partition, one-to-one.

Worked example ✍ R = {(1,1),(2,2),(3,3),(1,2),(2,1)} on {1,2,3}:
Reflexive: three loops present. Symmetric: (1,2) and (2,1) both present. Transitive: (1,2),(2,1) forces (1,1) ✓ and (2,1),(1,2) forces (2,2) ✓. So R is an **equivalence relation** with classes **{1, 2}** and **{3}** — and it is **not** a partial order, since (1,2) and (2,1) with 1 ≠ 2 break antisymmetry.

Partial order (poset): **reflexive + antisymmetric + transitive**. Add "every two elements are **comparable**" and you get a **total order**. The Hasse diagram drops the loops and keeps only the **covering** arrows upward.

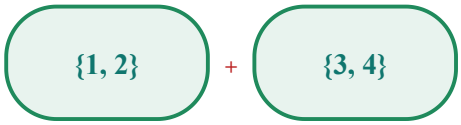
Worked example ✍ congruence mod 3 on ℤ: a R b iff 3 divides (a − b). Reflexive: a − a = 0 = 3·0. Symmetric: a − b = 3m ⇒ b − a = 3(−m). Transitive: 3m + 3k = 3(m + k). Classes [0] = {…, −6, 0, 3, 6, …}, [1] = {…, −2, 1, 4, 7, …}, [2] = {…, −1, 2, 5, 8, …} — disjoint, union ℤ.

Equivalence relations on a small set = Bell numbers:

They correspond one-to-one to the partitions, so the count is B_n: $B_1 = 1 \cdot B_2 = 2 \cdot B_3 = 5 \cdot B_4 = 15 \cdot B_5 = 52$. A 4-element set gives 15 = 1 (one block) + 7 (two) + 6 (three) + 1 (four).

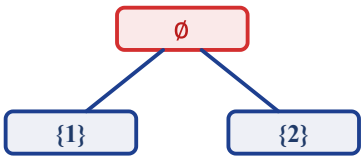
Memory trick: Both are named for what they do: **E**quivalence lumps equal-looking things into **classes**; a partial order gives **P**artial **O**rdering, where some things are ranked and others are not comparable. **⊆ on P(A) is always a partial order** — the standard example.

Exam trap: A relation is **both** an equivalence relation and a partial order only when every class is a singleton, i.e. R is the identity relation. And "not symmetric" does **not** make a relation antisymmetric — antisymmetry is a separate test, not a negation.



a partition of a 4-element set
⇒ one equivalence relation

classes never overlap and never leave a gap — that is exactly what "equivalence" means



P({1,2}): a poset

loops and the arrows ∅ → {1,2} are omitted; {1} and {2} are **not comparable**

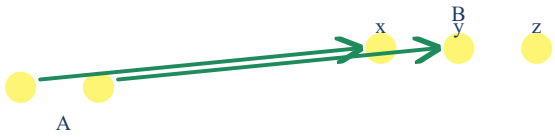
8 Functions: Domain, Codomain, Range

Definition: A **function** f: A → B is a relation from A to B in which **every element of A appears exactly once as the first component of a pair**. Equivalently, $(a_1, b_1), (a_2, b_2) \in R$ with $a_1 = a_2$ force $b_1 = b_2$. Nothing is said about how often an element of B is hit.

Name	It is	In code
Domain	the whole set A the function is declared on	every x that must be handled
Codomain	the declared target set B	the set the signature promises
Range / image	f(A) = { f(a) | a ∈ A } ⊆ B — only the values actually taken	the values you can really observe
Constant function	one fixed value for every input; the range is a singleton	the same constant returned always
Total / partial	total = every domain element has an image; partial may not	no null

Injective, surjective, bijective

Name	Test	Plain words
Injective one-to-one	$f(a_1) = f(a_2) \Rightarrow a_1 = a_2$	no two inputs share an output
Surjective onto	$\forall\, b \in B\ \exists\, a \in A : f(a) = b$	no output is left unused
Bijective	both of the above	a perfect pairing



injective, NOT onto



onto, NOT injective

bijective needs: $|A| = |B|$

one green arrow per input — no output is hit twice, and none is skipped

The counting facts for finite sets: $|\mathbf{A}| \le |\mathbf{B}|$ is **necessary** for an injective A→B

$|\mathbf{A}| \ge |\mathbf{B}|$ is **necessary** for a surjective A→B

A bijection needs $|\mathbf{A}| = |\mathbf{B}|$, and then range = codomain. So if $|A| > |B|$ a function still exists but it **cannot** be injective; if $|A| < |B|$ it **cannot** be surjective.

Exam trap: **Injective and surjective are independent.** f: {1,2,3} → {x,y} must be onto but can never be one-to-one. A **constant** function is onto only when $|B| = 1$, and one-to-one only when $|A| = 1$.

9 Composition, Inverse, Identity — and Counting Functions

Composition: Let g: A → B and f: B → C. Then **$(f \circ g)(x) = f(g(x))$**, a function A → C. Read right to left: **g acts first, f afterwards**. The domain of f ∘ g is the domain of g. Composition is **associative** but **not commutative**.

Inverse and identity: The **identity** map $I_A(x) = x$ changes nothing. f: A → B is **invertible ⇔ bijective**; then f^{-1}: B → A exists and **$f^{-1} \circ f = I_A$ and $f \circ f^{-1} = I_B$**. Formally $f^{-1}(y) = x$ whenever $f(x) = y$.

Counting functions A → B with $|A| = m$, $|B| = n$:

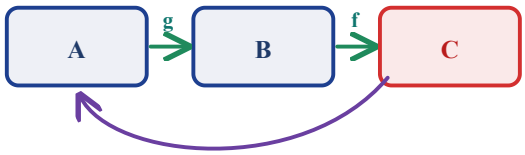
All functions: $\mathbf{n^m}$

Injective: $\mathbf{n(n-1)\cdots(n-m+1)} = n!/(n-m)!$

Surjective: $\mathbf{n!} \cdot \mathbf{S(m, n)}$, S = Stirling

Bijections: $\mathbf{n!}$ (only when m = n)

$(f \circ g)(x) = f(g(x))$



the purple arc is f^{-1} running back C → B

g first (green), then f — **read the composition right to left**

Memory trick: **∘ is a "then" sign: f ∘ g means "f after g"**, exactly as the brackets f(g(x)) say. The types forbid f ∘ g unless the codomain of g fits the domain of f.

Worked example ✍️ $|A| = 3, |B| = 4$: Each of the 3 inputs picks any of 4 outputs independently → $\mathbf{4^3 = 64}$ functions. ($24 = 4 \cdot 3 \cdot 2$ counts only the injective ones; $81 = 3^4$ would be functions $\mathbf{B \to A}$.) Surjective: 0, because 3 inputs cannot cover 4 outputs. Bijections: 0.

Worked example ✍️ inverse and composition: $f(x) = 2x + 3$. Let $y = 2x + 3 \Rightarrow x = \mathbf{(y - 3)/2}$, so $f^{-1}(x) = (x - 3)/2$, and $f((x - 3)/2) = x$ checks it out. Note $1/(2x + 3)$ is the **reciprocal**, never the inverse. For $f(x) = 2x + 1$ and $g(x) = x^2$: $(f \circ g)(x) = 2x^2 + 1$ but $(g \circ f)(x) = (2x + 1)^2$ — **not equal**.

Exam trap: (i) $(f \circ g)(x)$ is $\mathbf{f(g(x))}$, not $g(f(x))$ — check with a concrete x. (ii) Composition of bijections is bijective and injective ∘ injective is injective, but the mixed cases are not guaranteed. (iii) A relation has an inverse function only if it is a **bijection**, not merely injective.

Night-Before-the-Exam Sheet

Fact sheet:

$|A| = n \Rightarrow |P(A)| = \mathbf{2^n}$ subsets

of size k: $\mathbf{C(n,k)} = n!/k!(n-k)!$

$|A \cup B| = |A| + |B| - |A \cap B|$

disjoint $\Rightarrow |A \cup B| = |A| + |B|$

$A \cup A' = U \cdot A \cap A' = \emptyset$

$(A')' = A \cdot A - B = A \cap B'$

De Morgan: $(A \cup B)' = A' \cap B'$

$(A \cap B)' = A' \cup B'$

$|A \times B| = |A| \cdot |B|$

relations A→B = 2^(|A||B|)

on an n-set: $\mathbf{2^{(n^2)}}$ relations

$n = 3 \Rightarrow \mathbf{512}$

injective $\Rightarrow |A| \leq |B|$

surjective $\Rightarrow |A| \geq |B|$

bijection $\Rightarrow |A| = |B| \Leftrightarrow f^{-1}$ exists

A→B: $\mathbf{|B|^{|A|}}$ functions

$(f \circ g)(x) = f(g(x))$ — f after g

equivalence = R + S + T · poset = R + anti + T

bijections on an n-set: **n!**

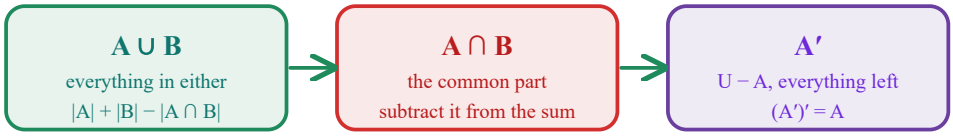
injective count: n!/(n−m)!

Top 12 one-liners for MCQs

1. 4 elements ⇒ **16** subsets, 15 proper
2. Sets ignore order and repeats
3. $|P(\emptyset)| = 1$, $|P(\{a\})| = 2$
4. $|A \cup B| = |A| + |B| - |A \cap B|$
5. $A \times B \neq B \times A$
6. 3 elements ⇒ $\mathbf{2^9 = 512}$ relations
7. Relation = **any** subset of $A \times B$
8. Symmetric ≠ antisymmetric
9. Empty relation: symmetric, anti, transitive
10. $A \to B = \mathbf{|B|^{|A|}}$ functions
11. f^{-1} of 2x+3 is $\mathbf{(x-3)/2}$
12. $f \circ g$ means **f after g**

Top traps: counting subsets as $2^n - 1$ without reading "proper" · treating **symmetric** and **antisymmetric** as opposites · writing 2^n relations on an n-set when it is 2^(n²) · "every relation is a function" · "injective implies surjective" · swapping f ∘ g into g(f(x)) · calling 1/(2x+3) an inverse

All tricks together: "A set is a basket with a label" · "in or out, n times, gives 2^n" · "⊆ is all-in, ⊂ is all-in and more outside" · "a relation is any subset of the grid" · "symmetric reverses the arrow, antisymmetric allows at most one" · "equivalence lumps equal things into classes" · "∘ is a then-sign: f after g" · "bijective is the only kind that can be undone"



the three operations the whole MCQ paper is built on

✓ Practice MCQs (exam pattern, 10 marks)

1. A set A has 4 elements. How many subsets?
(a) 8 (b) 16
(c) 4 (d) 32

2. A = {1,2,3}, B = {2,3,4}. A ∩ B is:
(a) {1,4} (b) {1,2,3,4}
(c) {2,3} (d) {1}

3. Relations on a 3-element set:
(a) 9 (b) 64
(c) 27 (d) 512

4. |A| = 3, |B| = 4. Functions A→B:
(a) 12 (b) 64
(c) 81 (d) 24

5. f(x) = 2x + 3. Its inverse is:
(a) (x+3)/2 (b) 2x − 3
(c) (x−3)/2 (d) 1/(2x+3)

6. Equivalence relations on a 4-element set:
(a) 16 (b) 52
(c) 15 (d) 24

7. {(1,2),(2,1)} on {1,2,3} is:
(a) symmetric only (b) antisymmetric
(c) both (d) neither

8. (A ∪ B)′ equals:
(a) A′ ∪ B′ (b) A′ ∩ B′
(c) A ∩ B (d) A − B

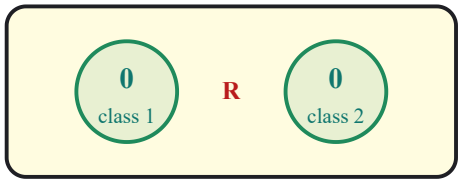
9. |A| = 5, |B| = 2. A function A→B must be:
(a) surjective (b) injective
(c) bijective (d) constant

10. Subsets of A = {1,2,3} of size 2:
(a) 2 (b) 3
(c) 6 (d) 8

Answer key → 1-b · 2-c · 3-d · 4-b · 5-c · 6-c · 7-a · 8-b · 9-a · 10-b

Written question: On the integers ℤ define a R b iff 3 divides (a − b). Prove R is an equivalence relation, list its classes, then give on A = {1, 2, 3} a relation that is reflexive and symmetric but not transitive. 8 marks

1. **Reflexive** — $a - a = 0 = 3 \cdot 0$, so 3 divides $a - a$ and $a \, R \, a$. 2
2. **Symmetric** — if $a - b = 3m$ then $b - a = 3(-m)$, and $-m$ is an integer, so $b \, R \, a$. 2
3. **Transitive** — $a - b = 3m$ and $b - c = 3k$ give $a - c = 3(m + k)$, so $a \, R \, c$; hence R is an **equivalence relation**. 2
4. **Classes** by remainder on division by 3: $[0] = \{\ldots, 0, 3, 6, \ldots\}$, $[1] = \{\ldots, 1, 4, 7, \ldots\}$, $[2] = \{\ldots, 2, 5, 8, \ldots\}$ (all negatives included) — disjoint, union $\mathbb{Z}$. 2



inside one class every pair is related — that is transitivity inside a class

Counter-example: On $\{1,2,3\}$ take $R = \{(1,1),(2,2),(3,3),(1,2),(2,1)\}$. Reflexive and symmetric, but **not** transitive: (1,2) and (2,3) would force (1,3), which is absent.

Before the exam ✓: 2^n, $C(n,k)$ and $|A \times B|$ instantly · all set laws with symbols · the six properties on a small relation · equivalence vs partial order.